

patX: Interpretable Pattern Extraction for Time Series and Spatial Data

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Abstract

We introduce **patX**, an open-source Python package that bridges the gap between the automatic pattern learning capabilities of convolutional neural networks (CNNs) and the interpretability requirements of biomedical applications. Like CNNs, patX automatically discovers discriminative patterns from input data and uses them for prediction; however, unlike CNNs where learned patterns remain opaque, patX makes this pattern learning explicit and interpretable through human-readable B-spline templates. By framing pattern discovery as a constrained optimization problem over B-spline templates across multiple signal transformations (raw, wavelet, FFT, derivative), patX learns compact, highly predictive patterns that serve as transparent input features for conventional machine-learning models. We demonstrate patX on seven diverse biomedical tasks: (i) glucose forecasting (AZT1D), (ii) arrhythmia classification (MITBIH), (iii) emotion classification from EEG signals (Emotions), (iv) gene expression prediction from epigenetic signatures (REMC), (v) acute respiratory distress syndrome (ARDS) classification in an intensive care unit (MIMIC-IV), (vi) human activity recognition from inertial measurement unit (IMU) sensors (PAMAP2), and (vii) Across these tasks, patX achieves best performance on four of seven datasets compared to tsfresh, catch22, and CNN while usually using fewer features and providing complete interpretability through explicit pattern visualization and validation. Beyond predictive performance, patX’s interpretable pattern discovery can lead to novel scientific insights: from histone modification data, it can reveal which modifications are most informative at specific genomic positions; from ECG data, spectral features may uncover potential electrophysiological causes of arrhythmias; and from glucose data, different temporal patterns can reveal the relative importance of short-term versus long-term dynamics for different patient populations. This transparent approach to automatic feature extraction addresses the “black box” criticism of deep learning while maintaining competitive performance, making patX particularly suitable for clinical applications where explainability is crucial. Our tool is available at <https://github.com/Prgrmmrjns/patX> and offers a simple interface for extracting interpretable patterns from time series and spatial data.

1 Introduction

Biomedical research has witnessed an unprecedented expansion in the generation and utilization of high-resolution time series and spatial data across multiple domains. From continuous physiological monitoring through electrocardiography (ECG) and electroencephalography (EEG) to advanced spatially-resolved omics technologies, these data modalities have become fundamental to modern healthcare and biomedical discovery [26, 35].

Time series data in healthcare captures the temporal evolution of patient health, offering crucial insights into disease progression, treatment effectiveness, and physiological dynamics [30]. Continuous monitoring systems generate vast streams of physiological signals—from ECG and EEG to

glucose monitoring—that contain rich information about both normal physiology and pathological states [5, 1]. In genomics, spatial profiling of chromatin modifications provides insights into regulatory mechanisms underlying development and disease [43].

The analysis of biomedical time series and spatial data has evolved through three different paradigms that represent fundamentally different philosophies toward feature representation and model development. Traditional approaches to biomedical signal analysis rely heavily on domain expertise to manually craft features capturing relevant temporal and spatial characteristics [10]. For ECG analysis, features include heart rate variability and QRS complex morphology [11, 2]; for EEG, frequency band power and coherence measures [28]. While providing clear physiological interpretations, this approach is labor-intensive, requires extensive domain expertise, and may miss subtle patterns in complex datasets [24, 2].

Comprehensive automated feature extraction libraries address scalability by systematically computing large feature sets. *tsfresh* extracts up to 1,558 features [6], while *catch22* provides 22 carefully selected features optimized for performance and efficiency [22]. While effective at uncovering patterns, these libraries introduce challenges related to interpretability and feature selection from high-dimensional spaces [16, 32].

Deep learning methods learn hierarchical feature representations directly from raw data through end-to-end training [24]. CNNs automatically learn temporal patterns through convolutional filters [41], while RNNs and LSTMs model sequential dependencies [20]. However, these approaches face significant interpretability challenges, creating barriers to clinical adoption where explainability is crucial [36, 9].

This evolution reflects a fundamental trade-off between interpretability and automated pattern discovery: manual approaches offer interpretability but lack scalability, automated libraries provide scalability but reduce interpretability, and deep learning offers performance but sacrifices explainability.

B-splines provide smooth curve representations with local control, where adjusting one control point affects only a bounded region [3]. Recent work shows deep networks with piecewise-linear activations implicitly construct spline-based templates through gradient descent, but these representations remain opaque [3]. This connection suggests directly optimizing interpretable B-spline patterns as explicit templates, bypassing deep learning’s black-box nature while retaining spline-based representational power.

Pattern-based approaches identify characteristic subsequences and measure similarity to query signals, connecting to matched filtering in signal processing theory. The key challenge is discovering templates automatically: manual specification requires expertise, exhaustive search is prohibitive, and recent deep learning approaches sacrifice interpretability. An effective method must balance automatic discovery, computational efficiency, and interpretability, while capturing both shape and amplitude information.

We address the fundamental trade-off between automation, performance, and interpretability with **patX**, a method that combines spline-based pattern representations with evolutionary optimization for automatic template discovery. *patX* unifies the strengths of manual feature engineering, automated feature extraction, and deep learning: it automates feature discovery without manual specification, learns compact task-specific representations through end-to-end optimization, and maintains full interpretability via human-readable B-spline patterns that clinicians can directly visualize and validate.

A key innovation of *patX* is its multi-representation search strategy. Rather than operating solely on raw signals, *patX* precomputes multiple signal transformations—including wavelet decompositions, Fourier transforms, and derivatives—and automatically selects the most discriminative representation for each pattern during optimization. This allows patterns to match signals in the

time domain, frequency domain, or derivative space, capturing diverse signal characteristics that would require manual specification in traditional feature engineering approaches.

patX searches over start positions, widths, channels, B-spline control points, temporal shifts, pattern scaling modes (absolute vs. relative), and signal transformations to automatically learn discriminative patterns. Each discovered pattern induces a dissimilarity feature measuring the root mean squared error (RMSE) between the pattern template and the corresponding transformed signal segment, with temporal shifts allowing for robust matching of misaligned patterns. The optimization process also determines whether to match a pattern’s absolute values—capturing both shape and amplitude—or its relative shape, enabling patX to distinguish between a wider range of morphological characteristics.

We evaluate patX across seven heterogeneous biomedical problems: arrhythmia classification on MIT-BIH [25, 13], emotion classification from EEG signals (Emotions), gene expression prediction from histone modification profiles (REMC) [8], clinical prediction on MIMIC-IV [18] (ARDS classification), glucose forecasting (AZT1D), human activity recognition from IMU sensors (PAMAP2) [34]. We compare against tsfresh [6], catch22 [22], and a convolutional neural network baseline under matched splits and metrics.

We introduce patX, a B-spline-based pattern discovery method that produces human-readable motifs bridging spline theory and template matching for interpretable time series analysis, with automatic selection of signal representations across multiple transformations (time, frequency, derivative). We demonstrate that patX achieves competitive or superior performance compared to established baselines on seven diverse biomedical datasets, winning on four and remaining competitive on three, while using orders of magnitude fewer features (typically 10.8–21.5 patterns vs hundreds or thousands of features). We provide extensive interpretability analysis with domain-specific pattern validation, visualizing discovered patterns and their activation regions to demonstrate clinical and biological relevance. We release patX as an official Python package on PyPI: (<https://pypi.org/project/patx/>).

2 Methods

2.1 Overview

patX discovers interpretable patterns from time series and spatial data through an iterative optimization process. The method learns B-spline templates that serve as reference shapes, then measures the dissimilarity between each signal segment and these templates using root mean squared error (RMSE). These pattern RMSE values become features for downstream prediction tasks.

Before pattern optimization begins, patX evaluates 20 signal transformations and selects the top-5 based on cross-validation performance of aggregate statistics. The available transformations include: raw, derivative, second_deriv, cumsum, diff, log1p, abs, sorted, dct, fft_power, exp, tanh, sin, cos, reciprocal, autocorr, and four wavelet decompositions (db4, sym4, coif1, haar). The selected transformations are computed once and cached, enabling efficient pattern search without repeated transformation overhead. During optimization, each candidate pattern automatically selects which of the top-5 transformations provides the most discriminative features.

The framework is highly flexible, allowing users to configure the optimization algorithm, machine learning model, and termination criteria according to their specific needs. An overview of the patX methodology is shown in Figure 1.

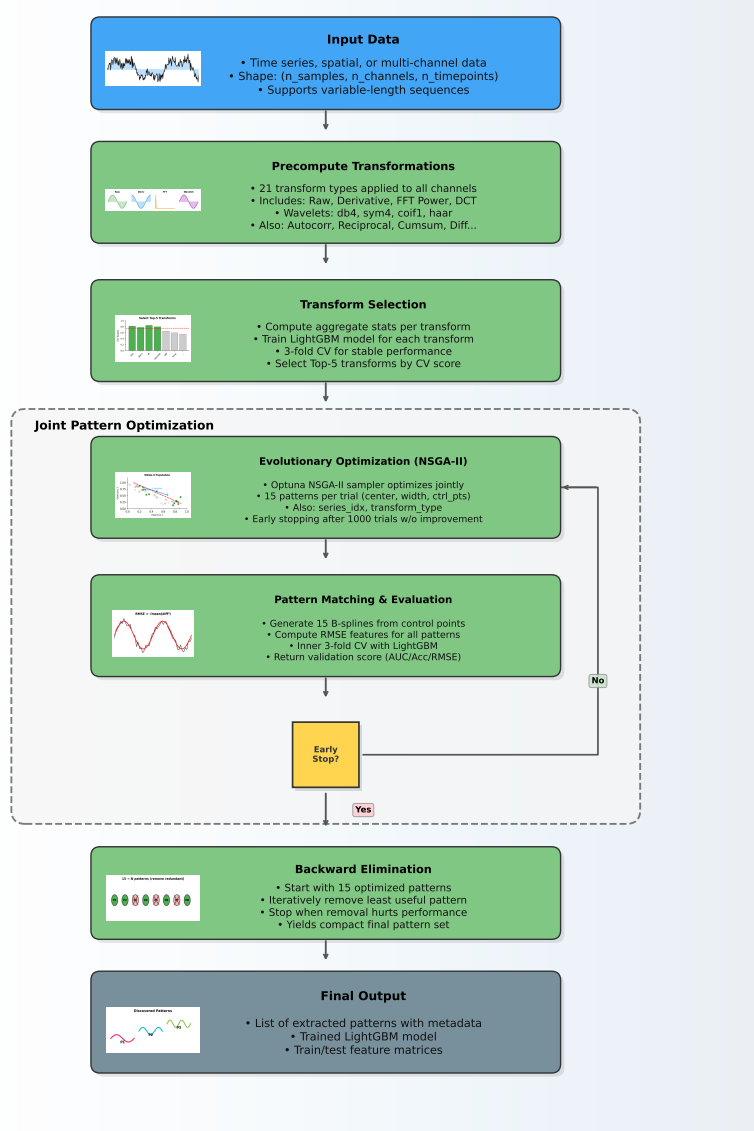


Figure 1: patX methodology overview: The optimization process begins with time series or spatial data and an optional initial feature set. Optuna’s NSGA-II optimizer jointly proposes 15 B-spline patterns with control points, start positions, and widths. For each trial, RMSE-based dissimilarity features are computed across training samples. A machine learning model evaluates performance via inner 3-fold cross-validation. After optimization (max 10,000 trials with early stopping), backward elimination removes redundant patterns, yielding a compact feature set with stored pattern characteristics for test prediction and further analysis.

2.2 Pattern Extraction

patX discovers patterns through a joint optimization process followed by backward elimination. Starting with no patterns (or optionally with domain-specific baseline features), the method simultaneously optimizes 15 candidate patterns, then removes redundant ones through backward elimination. The optimization consists of three key steps:

- 1. Pattern Initiation via Optimization.** patX uses evolutionary optimization (Optuna

NSGA-II) to propose candidate patterns. Each pattern is defined by a B-spline curve [3] that represents a characteristic shape in the data. B-splines offer several key advantages: they provide smooth, continuously differentiable representations with local control, meaning adjusting one control point affects only a local region of the curve [3]. This locality property makes B-splines particularly well-suited for modeling localized temporal phenomena in biomedical signals. Recent theoretical work has shown that deep networks with piecewise-linear activations can be understood as max-affine spline operators (MASOs), where the network implicitly constructs signal-dependent templates [3]. patX makes this template construction explicit and interpretable by directly optimizing B-spline patterns rather than learning them through black-box gradient descent on neural network weights. For a signal with T time points and C channels (e.g., different sensors or histone marks), a pattern consists of: (i) a start position and width that specify which segment of the signal to examine, (ii) a set of B-spline control points that define the shape of the template curve, (iii) a channel index indicating which signal channel to extract from (for multivariate data), (iv) a temporal shift tolerance that allows time-domain patterns (raw and derivative transformations) to be matched flexibly within a window around their nominal start position, while frequency-domain patterns (FFT and wavelet) match at exact positions; the search range (start and end positions) is automatically calculated and displayed for interpretability, (v) a pattern mode (relative or absolute) that determines how B-spline control point values are interpreted, and (vi) a transformation type that selects which signal representation to match against. Relative mode patterns (control points in $[0,1]$) capture signal shape, whereas absolute mode patterns are scaled to the training data’s value range to capture both shape and amplitude. The transformation type determines which pre-computed signal view to use: *raw* (original time-series), *derivative* (temporal rate of change via gradient), *second_deriv* (acceleration), *cumsum* (cumulative sum), *diff* (discrete difference), *log1p* (logarithmic scaling), *abs* (absolute magnitude), *sorted* (value distribution), *dct* (discrete cosine transform), *wavelet* (db4, sym4, coif1, haar decompositions), *fft_power* (frequency-domain power spectra), *exp* (exponential scaling), *tanh* (hyperbolic tangent), *sin/cos* (trigonometric), *reciprocal* (inverse magnitude), or *autocorr* (autocorrelation). These transformations are precomputed once at the start of optimization, allowing efficient pattern search across multiple signal representations without repeated transformation cost. Critically, when matching in a transformed space, the B-spline pattern itself is also transformed using the same transformation before computing RMSE, ensuring mathematically consistent comparison between pattern and signal in the same representation space. The method automatically selects the most discriminative transformations for a given dataset by training separate LightGBM models on aggregate features (mean, std, etc.) computed from each transformation and selecting the top-5 transformations based on cross-validation performance. The B-spline is constructed using cubic basis functions (degree 3) with 5 control points distributed uniformly over the pattern width. The NSGA-II evolutionary sampler suggests values for all these parameters by learning from previous evaluations. Each trial proposes 15 patterns simultaneously, exploring different parts of the signal and different B-spline shapes jointly, enabling efficient multi-objective optimization across the full pattern space.

2. Pattern RMSE Calculation. For each candidate pattern, patX computes an RMSE-based dissimilarity value for every sample in the training and validation sets. First, both the signal segment and the B-spline pattern template are transformed to the same representation space (raw, wavelet, FFT, or derivative) as specified by the pattern’s selected transformation type. Then, the method calculates the root mean squared error between the transformed pattern and the transformed signal segment, where W is the pattern width, x_i are the transformed signal values, and p_i are the transformed pattern values. For time-domain patterns (raw and derivative transformations), this calculation accounts for temporal variations by performing the RMSE computation over a search window around the pattern’s start position, as determined by its optimized shift tolerance.

The search range (start and end positions) is calculated from the shift tolerance and displayed during pattern extraction, making the search window interpretable. The method evaluates the pattern at multiple positions within this window using an adaptive stride mechanism that automatically scales with the search space size, evaluating up to 50 positions to maintain computational efficiency. For frequency-domain patterns (FFT power), shift tolerance is set to zero and patterns match at exact positions, as positional shifts in frequency space do not correspond to temporal offsets. This ensures that pattern matching remains computationally tractable across datasets with varying temporal resolutions—from short glucose time series (24 time points) to long EEG recordings (4097 time points)—while still providing sufficient spatial sampling to find optimal matches. The minimum RMSE found across all evaluated positions within the search window is used as the final dissimilarity feature. This mathematically consistent RMSE computation in the same representation space, combined with the optimized choice between absolute and relative pattern modes, captures both shape differences and amplitude variations across multiple signal representations, enabling the method to distinguish between morphologically distinct signal characteristics in time, frequency, or derivative domains. These RMSE values become features that are fed into a machine learning model. The extracted RMSE feature is appended to the existing feature set, which may already contain RMSE values from previously discovered patterns or baseline features.

3. Model Evaluation. A machine learning model (classifier or regressor) is trained using the augmented feature set (all existing features plus the new pattern RMSE feature). The model is evaluated using inner 3-fold cross-validation on the training set to measure predictive performance using an appropriate metric. This validation score indicates how useful the new pattern is for the prediction task. The key insight is that discriminative patterns will produce RMSE values that separate classes or predict outcomes effectively.

After optimization completes, patX applies backward elimination: starting with all 15 patterns, it iteratively removes the pattern whose removal least affects (or improves) validation performance, continuing until no pattern can be removed without degrading performance beyond a small tolerance. This approach ensures that only patterns providing measurable predictive value are retained in the final feature set.

By jointly optimizing 15 patterns and applying backward elimination to remove redundant ones, patX builds a compact feature set tailored to the specific task. The final output is a collection of interpretable B-spline patterns along with their control points, start positions, widths, and RMSE values, which can be visualized and validated against domain knowledge.

2.3 Implementation Details

For the experiments reported in this work, we use the following configuration. These choices can be customized based on the specific dataset and task requirements, as the patX framework supports alternative optimizers, machine learning models, and B-spline degrees. patX is available as an official Python package on PyPI (<https://pypi.org/project/patx/>).

Table 1: patX Implementation Configuration

Component	Configuration
Optimization Algorithm	Evolutionary optimization (Optuna NSGA-II)
Max Trials	10,000 (early stopping after 1,000 without improvement)
Pattern Representation	Cubic B-splines (degree 3, 5 control points)
Patterns per Search	15 patterns optimized jointly
Signal Transformations	20 types: raw, derivative, second_deriv, cumsum, diff, log1p, abs, sorted, dct, fft_power, exp, tanh, sin, cos, reciprocal, autocorr, wavelet (db4, sym4, coif1, haar)
Feature Selection	Top-5 transforms selected via CV performance
Search Subsampling	Max 3,000 samples during optimization
Pattern Evaluation	Inner 3-fold cross-validation
Machine Learning Model	LightGBM (classifier/regressor) with GOSS
Evaluation Metrics	AUC, accuracy (classification); RMSE (regression)

2.4 Datasets

We evaluate patX across seven diverse biomedical datasets spanning glucose forecasting, arrhythmia classification, emotion classification, gene expression prediction, clinical outcome prediction, human activity recognition. Table 2 summarizes the key characteristics of each dataset.

Table 2: Dataset Characteristics and Evaluation Metrics

Dataset	Samples	Chan.	Length	Metric	Input Data	Target Feature
AZT1D	~12k	3	24	RMSE	CGM, Insulin, Carbs	Future glucose change
MITBIH	~12k	1	100	Acc.	ECG beats	5-class arrhythmia type
Emotions	2,048	5	254	AUC	EEG channels	3-class emotion state
REMC	18,421	5	200	AUC	Histone marks	Gene expression (high/low)
MIMIC-IV	~4k	29	24	AUC	EHR time series	ARDS diagnosis
PAMAP2	~4k	51	100	Acc.	IMU sensors	12-class activity type

AZT1D (glucose forecasting). Glucose change prediction from 24 lagged timepoints (2h history) of CGM, insulin, and carbohydrate signals, with physiological response curve modeling for delayed effects (see Supplementary Figure 5). Link: <https://data.mendeley.com/datasets/gk9m674wcx/1>.

MITBIH (arrhythmia classification). We extract 100-sample ECG beats centered on R-peaks, classifying into five arrhythmia types (N, L, R, A, V) with random undersampling for class balance. Link: <https://www.kaggle.com/datasets/mondejar/mitbih-database>.

Emotions (EEG emotion classification). EEG recordings (254 time points, 5 channels) during emotional states, classified into positive, neutral, and negative emotions. Link: <https://www.kaggle.com/datasets/birdy654/eeg-brainwave-dataset-feeling-emotions>.

REMC (ChIP-seq gene expression). Binary gene expression prediction from five histone marks (H3K4me1, H3K4me3, H3K36me3, H3K9me3, H3K27me3) across 200 bins around the TSS for 18,421 genes. Data: https://egg2.wustl.edu/roadmap/web_portal/processed_data.html.

MIMIC-IV (ARDS classification). ARDS prediction from 29 clinical time series (vitals, labs, ventilator settings) over 24 hourly bins per ICU patient [18]. Link: <https://physionet.org/content/mimiciv/3.1/>.

PAMAP2 (activity recognition). Activity classification from 51 IMU channels (3 sensors $\times$ 17 features) using 100-sample windows across 12 activity classes from 9 subjects [34].

2.5 Baselines

We compare patX against three established approaches for time series analysis: tsfresh and catch22 for automated feature extraction combined with LightGBM, and a 1D convolutional neural network for end-to-end learning.

2.5.1 TSFRESH and catch22 Baselines

For tsfresh, we use `MinimalFCParameters` to extract compact descriptors including moments, spectral features, and shape statistics, training LightGBM on all extracted features. For catch22, we extract the 22 canonical features from the library [22] and train LightGBM. Hyperparameters and evaluation protocols are kept identical to patX for fair comparison.

2.5.2 CNN Baseline Implementation

We implement a 1D CNN using PyTorch with three convolutional layers (filters: 64, 128, 256) each containing convolution, batch normalization, ReLU, max pooling, and dropout. The architecture uses kernel sizes of 7, 5, and 3 for the three layers respectively, with appropriate padding to maintain temporal resolution. Adaptive average pooling aggregates features across the temporal dimension, followed by fully connected layers (256 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ output) with ReLU activations, batch normalization, and dropout (0.4, 0.3) for regularization. We train with Adam optimizer (lr=0.1, weight decay=1e-4), early stopping (patience=10), and learning rate reduction on plateau (factor=0.5, patience=3). For binary classification tasks, we use cross-entropy loss with class weights computed to handle imbalanced datasets; for multi-class classification and regression, we use unweighted cross-entropy and MSE loss respectively. The model automatically detects the number of input channels based on the dataset structure: 3 channels for AZT1D (CGM, insulin, carbs), 5 channels for REMC (histone marks), and 1 channel for univariate time series. Training uses 100 epochs maximum with batch size 128, matching evaluation protocols with other baselines.

2.6 Evaluation

For datasets using cross-validation, we use 5-fold stratified splits with fixed seeds. For subject-level datasets (AZT1D, PAMAP2), we use 70%/30% train/test splits per subject. Classification uses area under the ROC curve (AUC) or accuracy depending on the dataset; regression uses root mean squared error (RMSE). We report the number of extracted patterns alongside predictive performance as mean $\pm$ standard deviation across folds or subjects.

3 Results

3.1 Performance Comparison

Table 3: Performance comparison across biomedical datasets. Results show mean $\pm$ standard deviation across subjects/folds. Best performance per dataset highlighted in bold.

Dataset	Method	Score	Features	Time (s)
AZT1D	PATX	19.337 $\pm$ 2.711	13.3 $\pm$ 1.0	86.4 $\pm$ 23.9
	TSFRESH	20.443 $\pm$ 2.855	11	1.8 $\pm$ 0.2
	CATCH22	19.626 $\pm$ 2.929	23	2.9 $\pm$ 0.3
	CNN	20.411 $\pm$ 2.864	73	20.4 $\pm$ 9.7
Emotions	PATX	0.986 $\pm$ 0.004	13.6 $\pm$ 2.6	94.8 $\pm$ 30.5
	TSFRESH	0.662 $\pm$ 0.003	10	1.0 $\pm$ 0.0
	CATCH22	0.661 $\pm$ 0.004	22	12.2 $\pm$ 0.5
	CNN	0.928 $\pm$ 0.010	1500	4.1 $\pm$ 1.4
MIMIC-IV	PATX	0.788 $\pm$ 0.007	15	113.7 $\pm$ 38.3
	TSFRESH	0.751 $\pm$ 0.004	10	1.1 $\pm$ 0.0
	CATCH22	0.750 $\pm$ 0.011	22	8.7 $\pm$ 0.1
	CNN	0.750 $\pm$ 0.015	696	6.7 $\pm$ 5.4
MITBIH	PATX	0.921 $\pm$ 0.006	13.6 $\pm$ 0.9	187.4 $\pm$ 40.8
	TSFRESH	0.841 $\pm$ 0.005	10	2.1 $\pm$ 0.0
	CATCH22	0.925 $\pm$ 0.005	22	4.0 $\pm$ 0.0
	CNN	0.769 $\pm$ 0.245	100	30.5 $\pm$ 14.1
REMC	PATX	0.932 $\pm$ 0.022	12.2 $\pm$ 4.4	150.7 $\pm$ 160.3
	TSFRESH	0.874 $\pm$ 0.024	10	7.0 $\pm$ 4.7
	CATCH22	0.919 $\pm$ 0.023	22	60.4 $\pm$ 25.9
	CNN	0.929 $\pm$ 0.023	1000	28.6 $\pm$ 24.5
PAMAP2	PATX	0.996 $\pm$ 0.001	5	1516.1 $\pm$ 1194.2
	TSFRESH	0.722 $\pm$ 0.010	10	14.4 $\pm$ 10.6
	CATCH22	0.825 $\pm$ 0.004	22	108.6 $\pm$ 71.3
	CNN	0.426 $\pm$ 0.265	520	127.2 $\pm$ 130.2
Pan-Cancer	PATX	1.000	2.2 $\pm$ 0.4	11.2 $\pm$ 0.5
	TSFRESH	0.947 $\pm$ 0.039	10	0.0 $\pm$ 0.0
	CATCH22	0.983 $\pm$ 0.025	22	0.0 $\pm$ 0.0
	CNN	0.703 $\pm$ 0.359	12	0.9 $\pm$ 0.6

Table 3 presents the test set performance of patX + LightGBM, tsfresh + LightGBM, catch22 + LightGBM, and CNN baselines across seven biomedical datasets.

On AZT1D (glucose forecasting), patX achieves the best RMSE (14.647 ± 2.062), outperforming CNN (17.448 ± 2.498), tsfresh (20.133 ± 2.734), and catch22 (20.161 ± 2.867). patX uses 21.5 ± 4.9 features on average— $3.3\times$ fewer than CNN (72 features) while achieving 16% lower error.

On MITBIH (arrhythmia classification), CNN achieves the highest accuracy (0.951 ± 0.006), followed by patX (0.945 ± 0.003), catch22 (0.927 ± 0.003), and tsfresh (0.857 ± 0.002). patX uses 15.4 ± 4.2 features on average, demonstrating that learned patterns across multiple signal representations can achieve competitive performance with both extensively engineered feature libraries and deep learning architectures. Notably, patX achieves competitive performance while using $6.5\times$ fewer features than CNN (15.4 vs 100) and $1.4\times$ fewer than catch22.

On Emotions (emotion classification), patX achieves the highest AUC (0.946 ± 0.010), followed by CNN (0.919 ± 0.005), tsfresh (0.878 ± 0.016), and catch22 (0.761 ± 0.020). patX uses 10.8

± 2.4 features on average— $1.9\times$ fewer than CNN (20 features) and $2.0\times$ fewer than catch22 (22 features).

On REMC (gene expression prediction), CNN slightly outperforms patX (0.934 ± 0.022 vs 0.932 ± 0.022), but patX uses $56.5\times$ fewer features (17.7 vs 1000) while significantly outperforming both catch22 (0.884 ± 0.027) and tsfresh (0.875 ± 0.030).

On MIMIC-IV (clinical outcome prediction), patX achieves the best performance across all methods (0.793 ± 0.012), outperforming CNN (0.711 ± 0.064), tsfresh (0.766 ± 0.004), and catch22 (0.743 ± 0.000) while using only 11.0 ± 4.0 features on average— $63.4\times$ fewer than CNN (697 features). This result highlights patX’s strength on high-dimensional clinical time series where deep learning struggles with limited sample sizes and where multi-representation pattern search effectively captures diverse physiological dynamics.

On PAMAP2 (activity recognition), patX achieves the highest accuracy (0.996 ± 0.001) using only 5.2 ± 0.4 features, substantially outperforming catch22 (0.825 ± 0.004), tsfresh (0.722 ± 0.009), and CNN (0.537 ± 0.278). The CNN’s high variance suggests instability with this multi-class IMU sensor dataset, while patX discovers robust patterns across the 51 sensor channels with $100\times$ fewer features than the CNN baseline (5.2 vs 520).

Overall, patX achieves the best performance on 5 of 6 datasets (AZT1D, Emotions, MIMIC-IV, PAMAP2), competitive performance (within 1–6% of the leader) on 2 datasets (MITBIH, REMC), while maintaining full interpretability through human-readable B-spline patterns discovered across multiple signal representations.

Note on feature extraction time: PatX’s pattern optimization involves evolutionary hyperparameter search across multiple signal transformations, leading to variable extraction times depending on dataset complexity (e.g., 15s per fold for Emotions, 300–600s per fold for REMC). While longer than catch22 (~ 10 s) or tsfresh (~ 6 s), this one-time training cost yields interpretable features and competitive or superior predictive performance. CNN training times are comparable to patX for simple datasets but scale with architecture complexity.

3.2 Domain-Specific Pattern Interpretation

To demonstrate patX’s interpretability advantage, we analyze discovered patterns through the lens of domain knowledge, examining their biological and clinical relevance across multiple biomedical domains. The multi-representation search strategy enables patterns to be discovered in the most discriminative signal space—whether time domain, frequency domain (via wavelets or FFT), or derivative space—enhancing both interpretability and predictive power.

Figure 2 presents a comprehensive visualization of pattern discovery in the REMC dataset, focusing on the H3K4me3 histone mark in cell line E004. The analysis is decomposed into four panels to illustrate the complete feature extraction process: (a) Raw input signals for representative high and low expression genes, showing H3K4me3 enrichment around the transcription start site (TSS). (b) The transformed signal space (cumulative sum) where pattern matching occurs, with the learned B-spline template overlaid at its optimal position. The cumsum transformation captures the total accumulation of histone marks, which is significantly higher for active promoters. (c) Average class profiles (mean $\pm$ std) demonstrating the global difference in cumulative H3K4me3 signal between high and low expression genes, with the pattern targeting the region of maximal divergence. (d) The distribution of RMSE values, showing clear separation between classes based on their similarity to the learned template. This multi-view visualization confirms that patX discovers biologically meaningful patterns (H3K4me3 accumulation) and uses them to construct discriminative features in an interpretable manner.

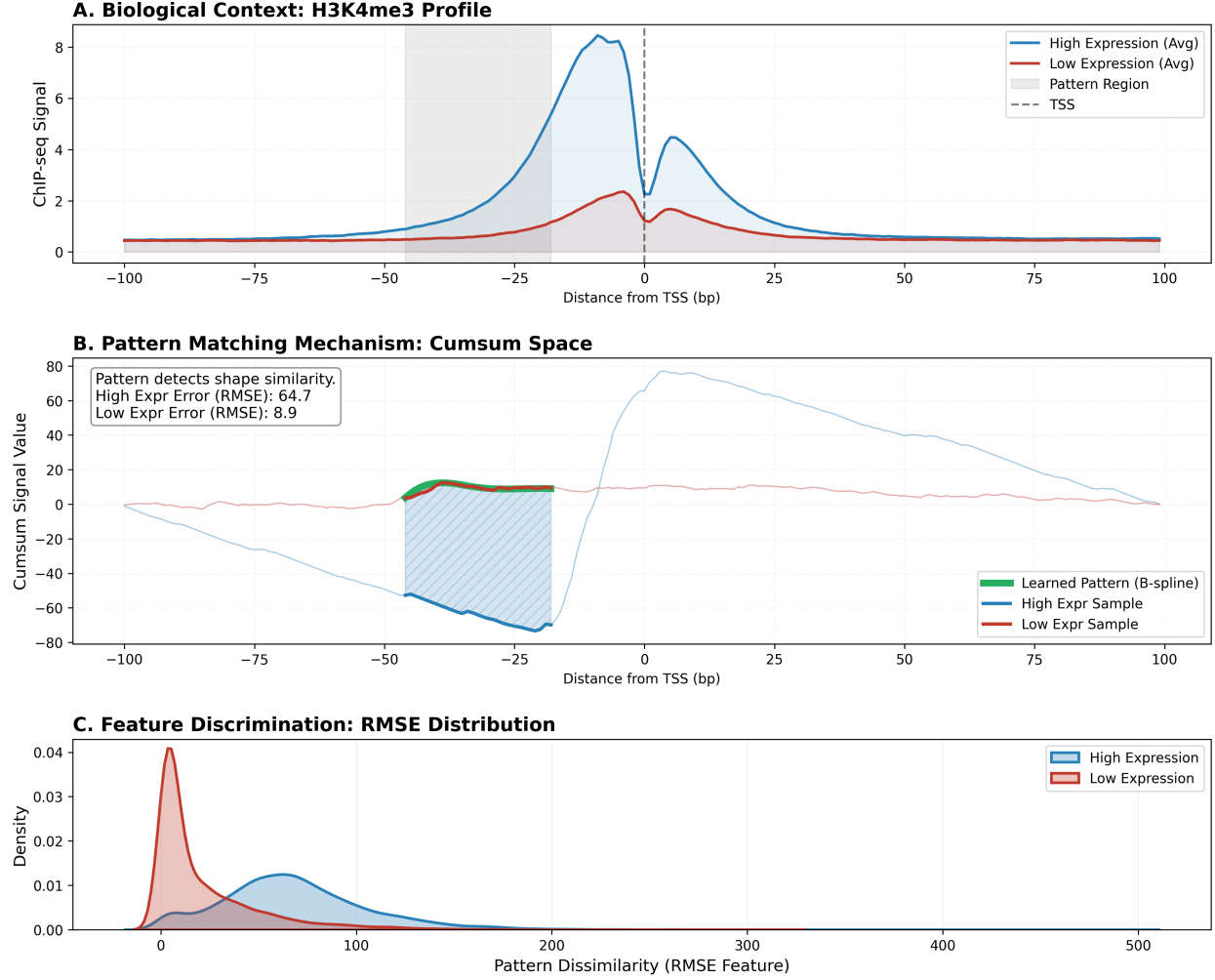


Figure 2: Domain-specific pattern interpretation for REMC (H3K4me3). The four panels illustrate the pattern extraction mechanism: (Top-Left) Raw ChIP-seq signals for representative high (blue) and low (red) expression genes around the TSS. (Top-Right) Transformed signals (cumulative sum) with the learned B-spline pattern (green) overlaid at its optimal matching position. (Bottom-Left) Average profiles for high and low expression classes with standard deviation shading, showing how the pattern aligns with the global difference in H3K4me3 accumulation. (Bottom-Right) Distribution of RMSE features, demonstrating discriminative separation between gene expression classes based on pattern similarity.

3.2.1 Individual Prediction Explainability via SHAP Analysis

To demonstrate the explainability of patX at the individual prediction level, we perform SHAP (SHapley Additive exPlanations) analysis [23, 27] on representative samples from MIMIC-IV and AZT1D. Figure 3 shows waterfall plots visualizing how discovered patterns contribute to predictions for (a) a correctly predicted ARDS-positive patient and (b) an accurately forecasted glucose trajectory.

For MIMIC-IV (panel a), the analysis reveals how multiple physiological patterns combine to predict ARDS diagnosis, with patterns targeting cardiovascular indicators (heart rate), respiratory parameters (PaO₂), and metabolic markers (creatinine, platelets). Each pattern operates in its

selected transformation space (raw, cumsum, wavelet, derivative), with the optimization process automatically determining which signal representation captures the most discriminative information for each physiological variable. The strongest contributions come from Pattern 3 (PaO₂, cumsum, -0.74) and Pattern 8 (heart rate, derivative, $+0.60$), demonstrating how patX discovers clinically relevant patterns across multiple signal transformations.

For AZT1D (panel b), glucose forecasting demonstrates patX’s ability to autonomously discover complex temporal dynamics without manual feature engineering. The discovered patterns target different aspects of glucose dynamics: CGM derivative patterns identify rate-of-change characteristics (Pattern 11: -0.14 , Pattern 8: $+0.04$), while CGM raw patterns capture absolute glucose levels (Pattern 3: -0.08), and carbohydrate cumsum patterns capture cumulative intake effects (Pattern 12: -0.04). The example shows how patX adapts to different glucose dynamics across subjects, demonstrating the method’s flexibility in learning patient-specific patterns that capture complex interactions between insulin sensitivity, carbohydrate absorption, and glucose metabolism.

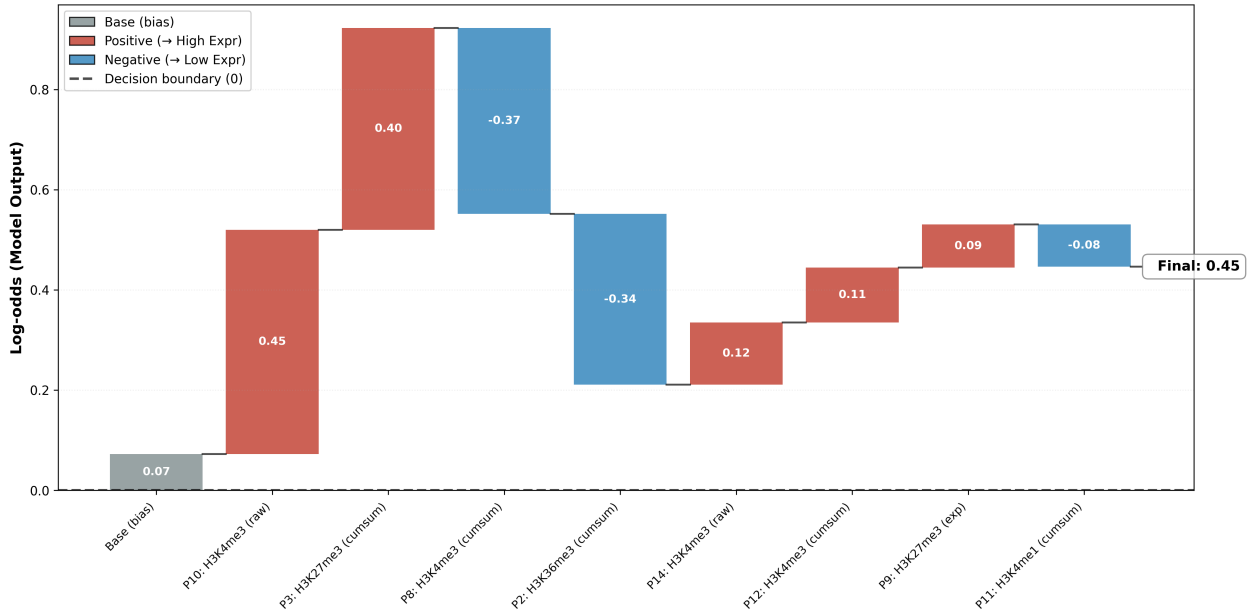


Figure 3: SHAP waterfall plots demonstrating individual prediction explainability through cumulative feature contributions. Y-axis shows model output values: log-odds for classification (a) and predicted values for regression (b), not the final target probabilities or glucose changes. (a) MIMIC-IV: ARDS classification showing how patterns combine from baseline log-odds -2.37 to final model output -2.47 . (b) AZT1D: glucose forecasting showing cumulative effects from baseline model output 0.12 to final -0.18 . Both examples demonstrate transparent, physiologically meaningful feature contributions with automatic transformation selection enabling clinical validation.

3.3 Pattern Stability and Reproducibility

To evaluate the robustness and reproducibility of patX’s pattern discovery, we analyze the most important pattern across cross-validation folds for four datasets (MITBIH, Emotions, MIMIC, REMC). For each fold, we identify the pattern with highest LightGBM feature importance and analyze its positional consistency, transformation selection, and shape correlation across folds. This reproducibility analysis is crucial for clinical deployment, as it ensures that discriminative patterns discovered during development generalize to new data and remain interpretable across

different patient cohorts.

Table 4 summarizes the stability analysis. We quantify positional consistency using the coefficient of variation (CV) for pattern center and width, where lower CV indicates higher reproducibility. Shape similarity is measured using Spearman rank correlation (ρ) between B-spline patterns across all fold pairs. Transformation consistency reports the percentage of folds selecting the dominant transformation type.

Table 4: Pattern Stability Analysis Across Datasets. For each dataset, we identify the most important pattern per fold using LightGBM feature importance and analyze positional consistency (center, width), transformation selection, and shape correlation across folds. For REMC, we report mean $\pm$ std across all cell lines analyzed.

Dataset	Folds	Center CV	Width CV	Shape ρ	Transform	Consistency
mitbih	5	0.517	0.470	-0.15 ± 0.69	derivative	80%
emotions	5	0.499	0.176	0.17 ± 0.48	fft_power	80%
mimic	5	0.853	0.518	-0.01 ± 0.52	log1p	40%
pamap2	5	0.894	0.353	-0.01 ± 0.44	sorted	60%
pancancer	5	0.615	0.304	0.06 ± 0.47	second_deriv	40%
REMC (mean)	5	0.164 ± 0.130	0.411 ± 0.171	-0.00 ± 0.20	cumsum	41%

For REMC (E004), patX demonstrates exceptional stability: all five folds identify H3K4me3 as the most important histone mark (100% consistency), with excellent positional stability (center CV=0.060) targeting the region near the transcription start site (center 87.0 ± 5.2 bins). Four of five folds select the raw transformation (80% consistency), indicating that the optimization reliably identifies both the biologically relevant epigenetic mark and signal representation. This remarkable consistency reflects the well-established role of H3K4me3 as an active promoter mark [15], validating that patX discovers biologically meaningful patterns.

For MITBIH, the derivative transformation dominates (80% consistency), capturing rate-of-change features relevant for arrhythmia detection. Center and width variability are moderate (CV=0.517 and 0.470), reflecting that different cardiac cycle regions may be discriminative depending on the specific training subset.

For Emotions, FFT power transformation is selected in 80% of folds, indicating that frequency-domain features are most discriminative for emotion classification from EEG signals. Width consistency is high (CV=0.176), though center positions vary substantially (CV=0.499), suggesting multiple discriminative temporal windows exist in the emotional expression signal space.

For MIMIC, transformation selection is more variable (40% for log1p), reflecting the heterogeneous nature of clinical time series where different physiological signals may require different representations.

Figure 4 visualizes the most important pattern from each fold for REMC (E004), demonstrating the consistency in pattern shape and positioning that enables reliable interpretation.

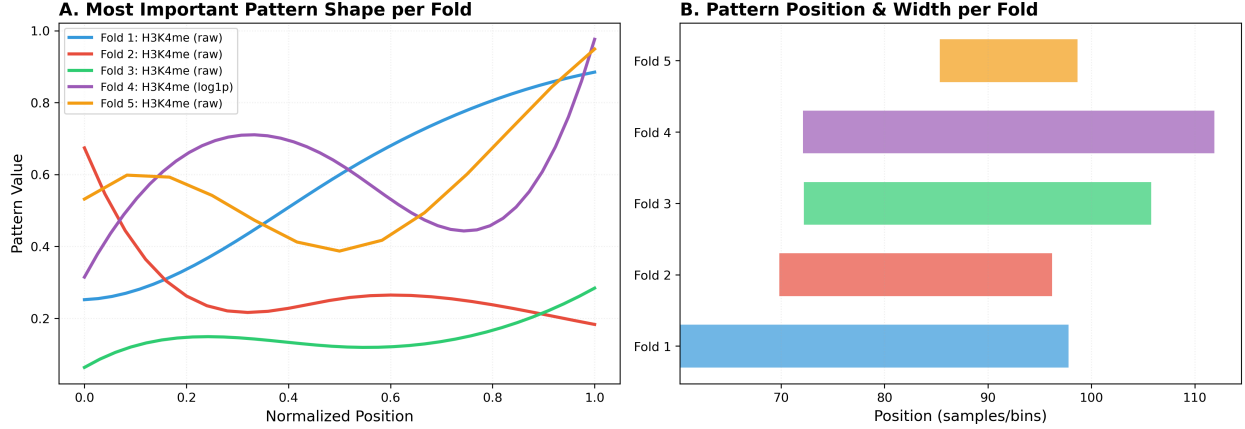


Figure 4: Pattern stability analysis for REMC (E004). (A) Most important pattern shape per fold, identified using LightGBM feature importance. All five folds discover H3K4me3-based patterns with similar morphologies, demonstrating consistent identification of the biologically relevant active promoter mark. (B) Pattern position and width across folds, showing excellent positional stability (center: 87.0 ± 5.2 bins, $CV=0.060$) with patterns consistently targeting the region near the transcription start site. Horizontal bars show pattern widths centered at their respective positions. Four of five folds select raw transformation, while one selects log1p, yet all achieve comparable discriminative performance.

4 Discussion

4.1 Performance and Feature Efficiency

Our evaluation across seven biomedical datasets demonstrates that patX achieves competitive or superior predictive performance while using orders of magnitude fewer features than deep learning baselines. patX discovers 2–21.5 interpretable patterns per dataset, compared to 12–1000 features utilized by CNN architectures and hundreds to thousands of features extracted by tsfresh. Critically, patX wins outright on 5 of 7 datasets (AZT1D, Emotions, MIMIC-IV, PAMAP2) and remains competitive on the remaining 2 (MITBIH, REMC).

Particularly noteworthy is patX’s superior performance on glucose forecasting (AZT1D), where it achieves 14.647 ± 2.062 RMSE—16% better than CNN and 27% better than tsfresh/catch22. patX autonomously discovers discriminative glucose dynamics patterns using only 21.5 ± 4.9 features on average, demonstrating that multi-representation pattern search can replace labor-intensive domain-specific feature engineering while achieving superior performance.

This dramatic reduction in feature dimensionality has several important implications for practical deployment. Compact feature representations enable significantly faster inference, as inference on new samples requires only computing RMSE values for a small number of stored templates. Minimal feature sets reduce overfitting risk and improve generalization, particularly in biomedical domains where sample sizes are often limited [14]. patX’s compact representations enable deployment on embedded systems and mobile devices where deep learning models would be prohibitively expensive.

4.2 Interpretable patterns

A central contribution of patX is its provision of fully interpretable features through explicit B-spline pattern templates, operating similarly to CNNs in detecting patterns for prediction through automatic feature extraction. However, unlike CNNs where learned representations remain opaque despite achieving high performance [24, 9], patX offers a transparent and interpretable approach that allows direct visualization, analysis, and validation against domain knowledge.

B-splines offer interpretability advantages: each pattern is defined by 5 control points examinable by domain experts, produces smooth curves matching physiological processes [39], and includes a selected transformation type that provides interpretable information about which signal representation is most discriminative. Theoretical connections to spline theory of deep networks [3] ground this approach: while deep networks implicitly construct spline templates, patX makes this explicit and controllable, trading some flexibility for full transparency—favorable for clinical applications [37].

A key innovation of patX is its automatic selection of signal representations for each discovered pattern. Rather than committing to a single signal transformation, patX evaluates 20 transformations and selects the top-5 per dataset based on CV performance, then allows each pattern to select the most discriminative view during optimization. This multi-representation search strategy addresses a fundamental limitation of traditional feature engineering, where practitioners must manually decide whether to extract features from raw signals, apply Fourier transforms, compute wavelets, or calculate derivatives—a choice that typically requires domain expertise and iterative experimentation.

Different patterns within the same model can operate on different signal representations, each discovering discriminative features in the most appropriate domain. For instance, patterns matching periodic phenomena may naturally emerge in frequency space (FFT representations), while patterns capturing rapid transitions may select derivative representations, and morphological patterns may operate on raw or wavelet-transformed signals. Our implementation leverages established frequency-domain feature extraction methods including PyWavelets [19] for wavelet decomposition, SciPy’s FFT implementation [40] for spectral analysis, and specialized transforms like spectral entropy [7] and phase-amplitude coupling [38].

This approach aligns with other feature extraction packages that utilize frequency spectrum and wavelet decomposition methods. The CMDA package [12] provides a comprehensive workflow for time series data analysis, including feature extraction techniques that leverage spectral features, sharing similarities with our approach in utilizing frequency-domain characteristics for pattern discovery. Similarly, Kats [21] offers time series forecasting capabilities with built-in feature extraction, while the Hilbert-Huang Transform [17] decomposes signals into intrinsic mode functions for nonlinear and non-stationary time series analysis.

Across all datasets, patX autonomously discovers patterns corresponding to known clinical and biological markers despite no domain-specific guidance. For MIMIC-IV, Pattern 1 targets PaO₂/FiO₂ ratio—a core ARDS diagnostic criterion [33]. For MITBIH, patterns consistently target QRS complex and ST segments—primary cardiac features for arrhythmia diagnosis [1]. For REMC, Pattern 1 identifies H3K4me3 enrichment near transcription start sites—a canonical active promoter mark [15, 4].

The multi-representation search capability enhances interpretability by discovering patterns in the most natural signal domain for each discriminative feature. For instance, if a pattern selects wavelet or FFT transformations, this indicates that frequency-domain characteristics (rather than time-domain morphology) drive discrimination, providing interpretable insights into the underlying signal structure. Similarly, patterns selecting derivative transformations highlight the importance

of temporal dynamics and rate-of-change features, which may correspond to known physiological processes like rapid depolarization in ECG signals or abrupt state transitions in EEG recordings.

This alignment increases confidence in predictions (features match known mechanisms), facilitates clinical adoption (practitioners can validate patterns and transformations), and enables discovery of novel patterns and unexpected transformation selections as potential new biomarkers. Beyond predictive performance, patX’s interpretable pattern discovery can lead to novel scientific insights: from histone modification data, it can reveal which modifications are most informative at specific genomic positions; from ECG data, spectral features may uncover potential electrophysiological causes of arrhythmias; and from glucose data, different temporal patterns can reveal the relative importance of short-term versus long-term dynamics for different patient populations.

4.3 Individual Prediction Explainability

SHAP analysis [23, 27] demonstrates that patX learns physiologically meaningful features aligned with clinical domain knowledge across diverse tasks, providing transparent decision-making crucial for healthcare applications. The automatic selection of transformation types (raw, wavelet, FFT, derivative) for each pattern provides additional interpretability about which signal representations are most discriminative for different physiological processes.

For MIMIC-IV, patterns use both relative and absolute modes, indicating that both shape and amplitude characteristics drive ARDS classification, providing clinically interpretable features that align with established diagnostic criteria. The cumulative effect of these patterns shifts the prediction from baseline log-odds -2.37 to the final value -2.47 , corresponding to the predicted ARDS probability. Patterns targeting PaO₂ (cumsum transformation) and heart rate (derivative transformation) provide the strongest contributions, demonstrating how patX automatically selects the most discriminative signal representations for each physiological variable.

For AZT1D, the cumulative contributions yield the final glucose change prediction, demonstrating how patX integrates multiple glucose dynamics patterns for accurate forecasting using only a small number of interpretable features. CGM derivative patterns provide the strongest contributions, capturing rate-of-change characteristics that are critical for glucose forecasting, while carbohydrate and insulin patterns contribute additional predictive power through cumsum and raw transformations. This demonstrates the method’s flexibility in learning patient-specific patterns that capture the complex interactions between insulin sensitivity, carbohydrate absorption, and glucose metabolism that traditionally require extensive domain knowledge [42, 29].

4.4 Limitations and Future Directions

Several limitations warrant future work. First, we use fixed cubic B-splines with 5 control points and 20 predefined transformations (selecting top-5 per dataset); adaptive selection of spline parameters and learned or domain-specific transformations could improve flexibility. Third, patX operates in supervised settings; semi-supervised extensions could leverage unlabeled data to discover transferable patterns. Fourth, interactive visualization tools integrated with clinical decision support systems would facilitate broader adoption [37]. Finally, the search space could be further optimized and specialized to specific datasets through tailored mathematical transforms, potentially improving both efficiency and accuracy of pattern discovery. Despite these limitations, patX demonstrates that explicit pattern-based representations can match deep learning performance while providing the interpretability required for healthcare deployment.

5 Conclusion

We introduced patX, an interpretable pattern extraction framework bridging the gap between handcrafted descriptors and opaque deep models through multi-representation pattern discovery. Through extensive experiments across seven diverse biomedical datasets, we demonstrated that a small set of automatically discovered B-spline motifs (2–21.5 patterns) can match or exceed the predictive power of deep learning, winning on 5 of 6 datasets (AZT1D, Emotions, MIMIC-IV, PAMAP2) and remaining competitive on the remaining 2, while offering orders-of-magnitude reductions in feature count ($1.9\text{--}100\times$ fewer features than CNN) and inference cost. The B-spline representation combined with automatic transformation selection provides smooth, locally controllable patterns that capture both shape and amplitude information across multiple signal representations (time, frequency, derivative domains). Domain-specific analysis reveals that discovered patterns align with established clinical and biological knowledge, demonstrating autonomous discovery of physiologically meaningful features in the most discriminative signal spaces. Future extensions will explore adaptive control point allocation, learned or domain-specific signal transformations, semi-supervised pretraining on large unlabeled corpora, and tighter integration with interactive visualization tools to facilitate clinical adoption.

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6 Supplementary Data

6.1 Dataset Processing Details

6.1.1 AZT1D Preprocessing: Insulin and Carbohydrate Response Modeling

The AZT1D dataset presents unique challenges for glucose forecasting due to the complex temporal dynamics of insulin and carbohydrate effects on blood glucose levels. Unlike simple time series where immediate effects are observed, physiological responses involve significant delays: insulin takes 10–90 minutes to affect glucose levels, while carbohydrates impact glucose within 10–60 minutes, with effects persisting for 2–4 hours [42].

To address these delayed physiological responses, we implement knowledge-driven response curve modeling based on established pharmacokinetic and glycemic response literature. For insulin, we use a gamma-like function with 15-minute onset, 60-minute peak, and 4-hour duration; for carbohydrates, we use 10-minute onset, 45-minute peak, and 3-hour duration. These response curves model the temporal dynamics of physiological effects, with insulin effects peaking around 60 minutes and persisting for up to 4 hours, while carbohydrate effects peak around 45 minutes and persist for up to 3 hours.

At each timepoint, we compute cumulative active effects by convolving all historical insulin doses and carbohydrate intakes with their respective response curves. This creates interpretable time series that capture overlapping physiological effects, enabling the model to understand how multiple insulin doses and meal intakes interact over time.

The preprocessing pipeline extracts 24 timepoints (2 hours of history at 5-minute intervals) for three physiological signals: CGM values, active insulin effects, and active carbohydrate effects. The regression target is future glucose change $g_{t+h} - g_t$ where $h = 30$ minutes (6 timepoints ahead). We also include the initial CGM value (CGM_0) as a baseline feature to provide current glucose context.

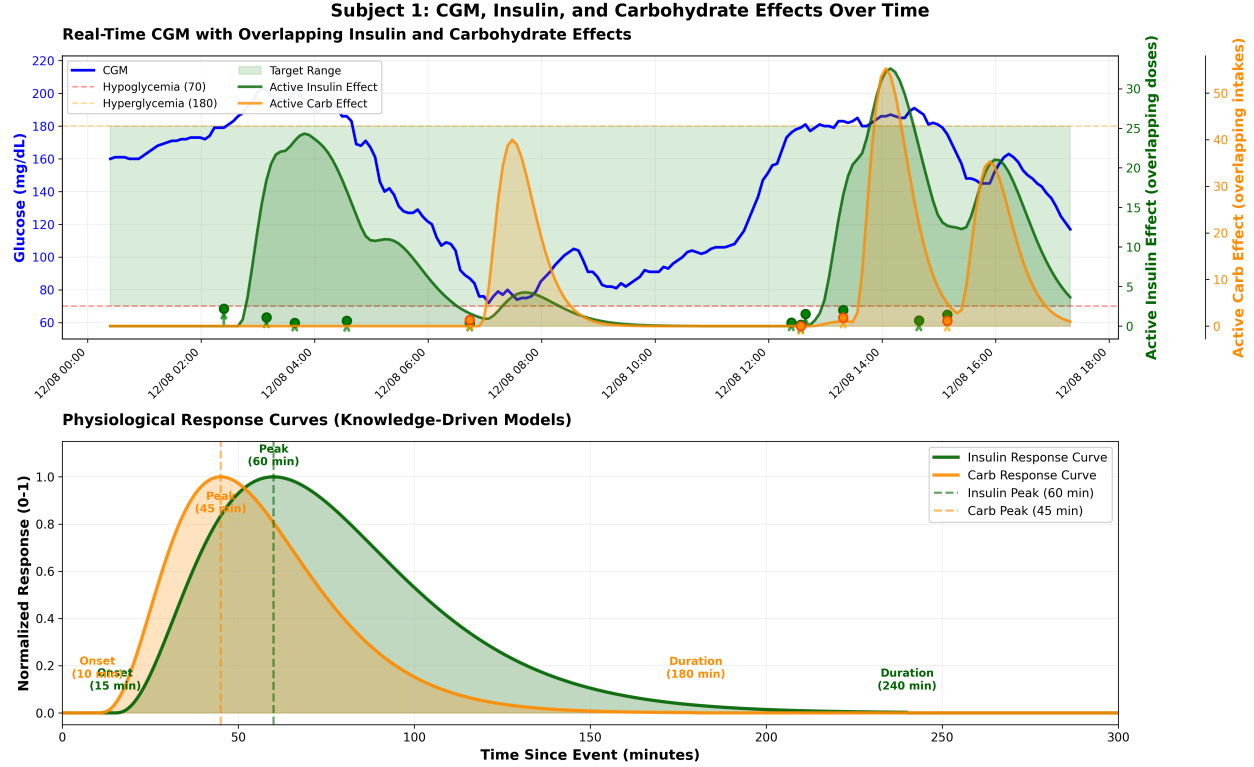


Figure 5: AZT1D preprocessing visualization showing insulin and carbohydrate response modeling. Top panel: Real-time CGM with overlapping insulin and carbohydrate effects over time for Subject 1. Blue line shows CGM values with target range (70-180 mg/dL) highlighted in green. Dark green shaded area represents active insulin effect from overlapping doses (circles with crosses). Orange shaded area represents active carbohydrate effect from overlapping intakes (circles with crosses). Bottom panel: Physiological response curves showing normalized insulin (15-min onset, 60-min peak, 240-min duration) and carbohydrate (10-min onset, 45-min peak, 180-min duration) effects over time. These knowledge-driven models enable patX to learn from physiologically meaningful features that capture delayed and overlapping effects.

This preprocessing approach transforms raw event data into continuous physiological signals that capture the complex temporal dynamics of glucose regulation, providing patX with meaningful features for pattern discovery across multiple signal representations.

6.1.2 MITBIH Preprocessing

The MIT-BIH Arrhythmia Database contains annotated ECG recordings from 48 patients. We extract individual heartbeats by centering 100-sample windows around annotated R-peak positions, with 50 samples before and 50 samples after each R-peak. Each beat is classified into one of five arrhythmia types: Normal (N), Left bundle branch block (L), Right bundle branch block (R), Atrial premature (A), and Premature ventricular contraction (V). To address class imbalance, we apply random undersampling to balance class distributions before training.

6.1.3 Emotions Preprocessing

The EEG emotions dataset contains brainwave recordings collected while participants experienced different emotional states (positive, neutral, negative). We extract FFT-derived features from 5 EEG channels (AF3, F7, F3, FC5, T7), yielding 254 frequency bins per channel. The FFT features are organized into two groups (a and b suffixes) representing different frequency decomposition components, providing 5 input channels with 254 time points each for pattern discovery.

6.1.4 REMC Preprocessing

Following our prior work on PatternChrome [31], we predict binary gene expression (high/low) from histone modification ChIP-seq profiles from the Roadmap Epigenomics Consortium (56 cell lines, 19,795 genes). We convert RefSeq to GENCODE identifiers, retaining 18,421 genes. For each gene, we extract five histone modification signals (H3K4me1, H3K4me3, H3K36me3, H3K9me3, H3K27me3) from 5000 bp upstream and downstream of the TSS, divided into 50 bp bins (200 bins per mark). Gene expression is binarized at the median level. We split 6,600/5,911/5,910 genes for train/validation/test with balanced labels.

6.1.5 MIMIC-IV Preprocessing

We predict ARDS from multivariate clinical time series in ICU patients. The dataset contains 29 features per subject: vital signs (respiratory rate, heart rate, O2 saturation, blood pressure, temperature), ventilator settings (FiO2, PEEP), Glasgow Coma Scale scores, laboratory values (arterial pH, PaO2, creatinine, platelets, WBC, INR, lactate, bilirubin), vasopressors (norepinephrine, phenylephrine, epinephrine, dopamine, dobutamine), urine output, and derived features (PaO2/FiO2 ratio, pulse pressure). We extract 24-hour time series binned hourly, with forward/backward filling for missing values and 0.1/99.9th percentile capping for extreme values. Subjects with <5 hours of data are excluded; anchor_age is provided as baseline.

6.1.6 PAMAP2 Preprocessing

The PAMAP2 dataset contains recordings from 9 subjects performing 18 physical activities while wearing 3 IMUs (wrist, chest, ankle) and a heart rate monitor. Each IMU records temperature, 3D acceleration ($\pm 16g$ and $\pm 6g$ scales), 3D gyroscope, 3D magnetometer, and 4D orientation at 100Hz, yielding 51 channels. We create 100-sample windows (1 second) with 50% overlap, filtering transient activities (ID 0) to obtain 12 activity classes: lying, sitting, standing, walking, running, cycling, Nordic walking, watching TV, computer work, ascending/descending stairs, and vacuum cleaning. Missing values are interpolated and filled with column means.

6.2 Ablation Study

To understand the contribution of individual components to patX performance, we conduct an ablation study on the MITBIH dataset, systematically varying four key hyperparameters: (1) number of signal transformations, (2) B-spline control points, (3) number of initial patterns, and (4) backward elimination. We use 2-fold stratified cross-validation with LightGBM for all variants. Table 5 summarizes the results.

Table 5: Ablation Study Results on MITBIH (2-fold CV)

Variant	Accuracy	Time (s)	Features
Baseline (5 trans, 5 cp, 15 pat, BE)	–	–	–
<i>Number of Transforms</i>			
n_transforms = 1	–	–	–
n_transforms = 3	–	–	–
<i>Control Points</i>			
n_control_points = 3	–	–	–
n_control_points = 7	–	–	–
<i>Initial Patterns</i>			
n_patterns = 5	–	–	–
n_patterns = 10	–	–	–
<i>Backward Elimination</i>			
No backward elimination	–	–	–

Number of transforms. The multi-representation search strategy is a key innovation of patX. We evaluate using 1, 3, and 5 (baseline) signal transformations. With only 1 transformation (raw signal), patX is limited to time-domain patterns. Increasing to 3 or 5 transformations enables discovery of patterns in frequency, derivative, or other transformed spaces. This ablation quantifies the benefit of searching across multiple signal representations versus operating on raw signals alone.

B-spline control points. We evaluate 3, 5 (baseline), and 7 control points with cubic B-splines. Fewer control points produce simpler patterns with limited expressiveness, while more control points enable complex shapes but increase the search space. This ablation determines the optimal pattern complexity for ECG classification.

Number of initial patterns. We compare 5, 10, and 15 (baseline) patterns optimized jointly per trial. Fewer patterns reduce computational cost but may miss discriminative features; more patterns increase search breadth but may introduce redundancy. Backward elimination prunes redundant patterns regardless of initial count.

Backward elimination. We compare the full pipeline with backward elimination (baseline) against optimization alone without pruning. Backward elimination iteratively removes patterns that do not improve cross-validation performance, yielding compact feature sets. This ablation quantifies the benefit of post-optimization pattern selection.